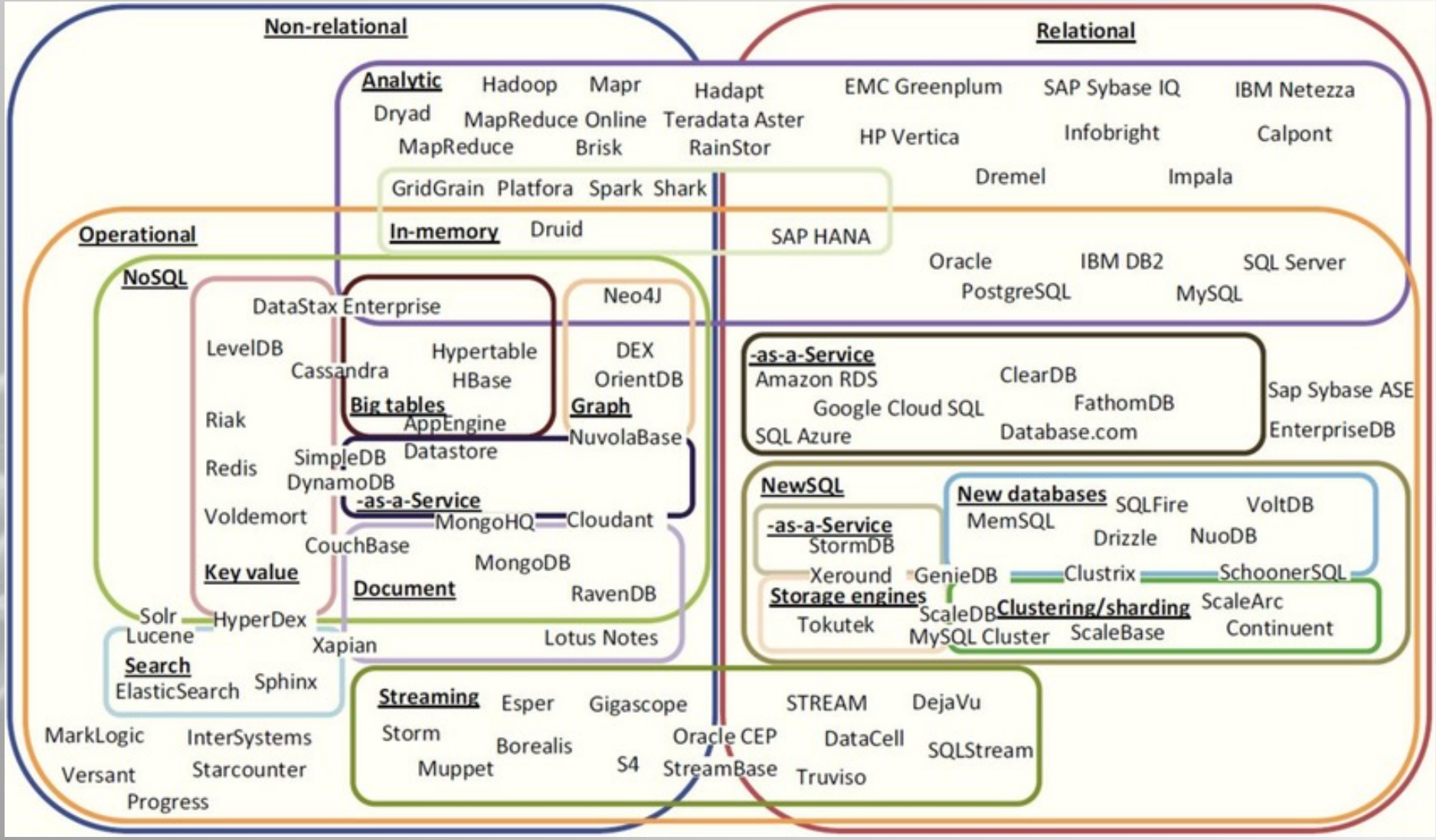
Databases

- Hierarchical Databases
  - root node & parent child relations
  - 1-to-1 or 1-to-n
- Network Databases
  - n-to-n
- Large Zoo of DB Types:
  - Relational Databases
    - uses structured query language SQL
  - Object Oriented Databases
  - Graph Databases
  - Document Databases
  - No-SQL

Simplified Schema for relational databases

- Snowflake Schema
  - higher normalization
- Star Schema
  - easier to understand

Data Base Zoo



Database options since the 2000's. Source: Korflatis et. al (2016)

Requirements on Database Systems

- ACID Compliance
  - Set of properties of an database
- CAP Theorem
  - Properties of an distributed data store

Good for decision making when DBMS needs to be selected

ETL & ELT

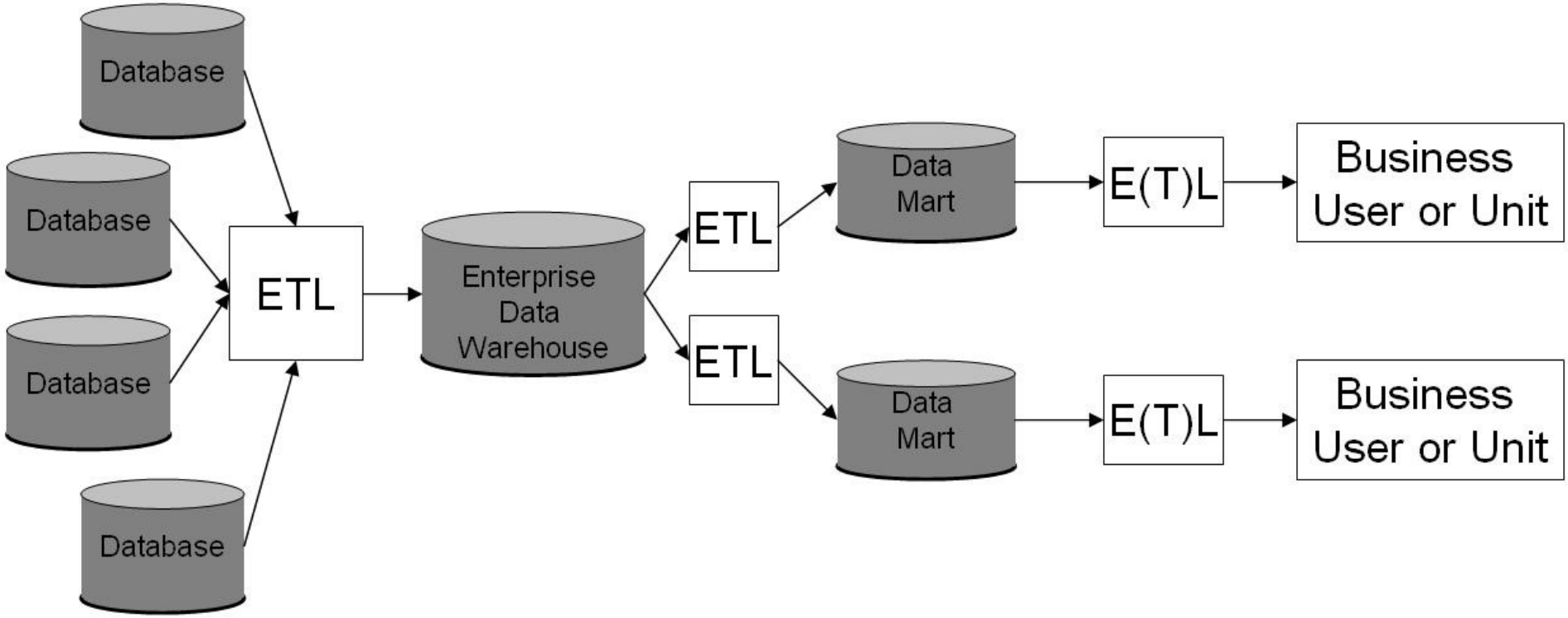
- Any possible order of
  - Extract, Transform, Load
- Important for building and using a Data Warehouse

Data Access in Systems

- Online Transaction Processing (OLTP)
  - Stable and DBMS are ACID compliant
  - Request often known before
  - Normalized data
  - Low volume
- Online Analytical Processing (OLAP)
  - Focus on analysis
  - Denormalized , optimized for analytics
  - Mostly read-only requests

Data Warehouse

Central data base with data from heterogenous sources.



By Jerry vlntn at English Wikibooks - Transferred from en.wikibooks to Commons., Public Domain, https://commons.wikimedia.org/w/index.php?curid=61795024

Data Warehouse

- “A Data Warehouse is a subject-oriented, integrated, non-volatile, and time variant collection of data in support of managements decisions” *(W. H. Inmon, Building the Data Warehouse, 2nd ed., John Wiley 1996)*

- Subject-Oriented: analyze a particular subject area, e.g. "sales"
- Integrated: data from multiple data sources
- Time-Variant: Historical data is kept, that contrasts with a transactions systems
- Non-volatile: data warehouse will not change, historical data should never be altered

DW more then a data base

Data Warehouse

- Integrate data
- OLAP (online analytics processing)
- Optimized for analytics queries
- Denormalized data

Data Base

- Real time data
- Very efficient
- OLTP (online transactional processing)
- Optimized for data modification queries
- Normalized data

Data Mart

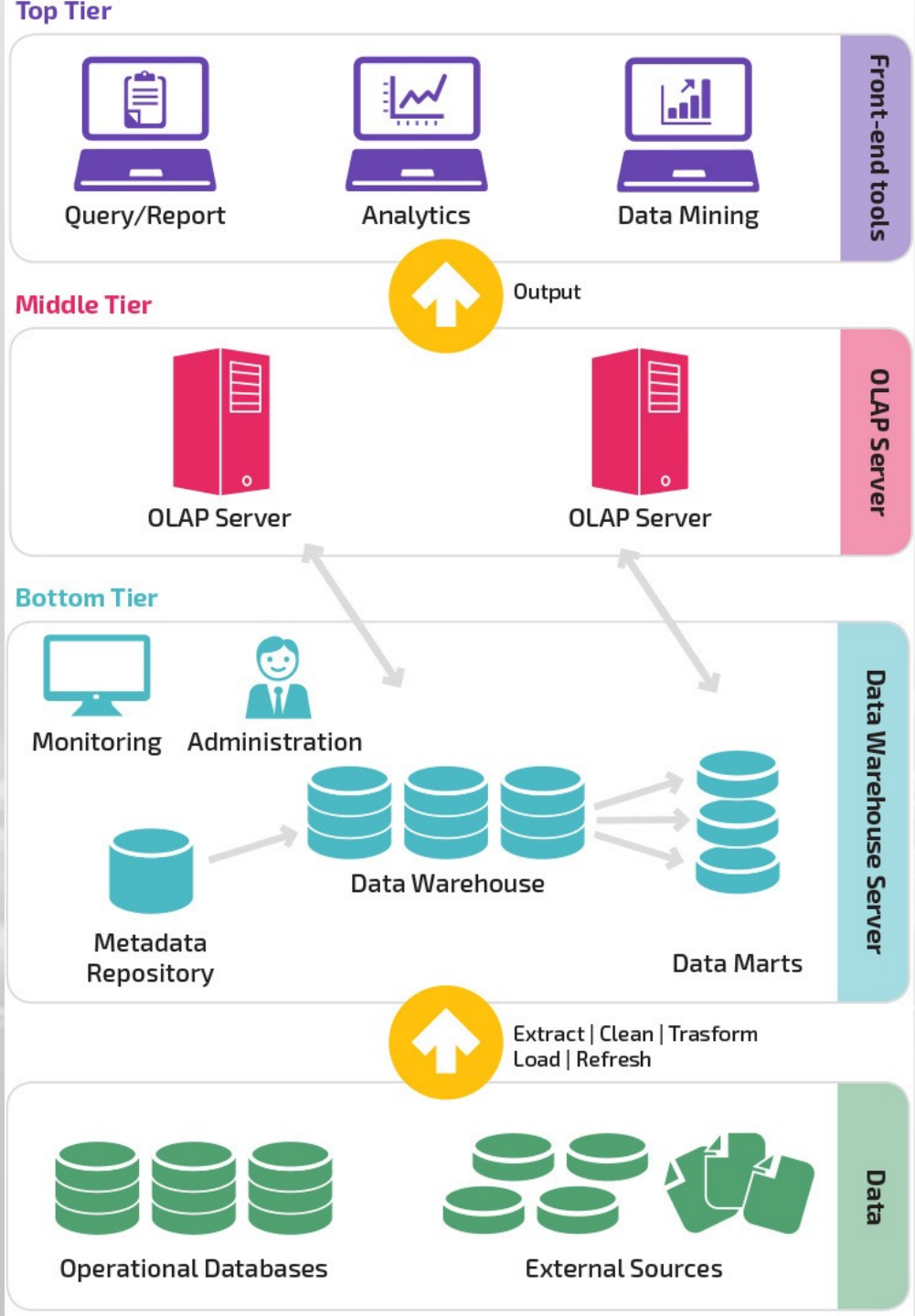
- a subject-oriented database designed
- to meet the needs of a specific group of users
- often a subsegment of an Enterprise Data Warehouse
- data access with higher performance then reading from the Data Warehouse
- can be created from the Data Warehouse or
- can be the building blocks for a new Data Warehouse

Advantages of Data Marts

- Data maintenance: different departments can take responsibility for and control their data
- Easy setup: the simple design requires less technical knowledge for implementation
- Analytics: key performance indicators (KPIs) can be easily tracked
- Easy input: data marts can serve as the foundation for a future enterprise data warehouse project

Service Layers

- Many components required for full pipeline



https://panoply.io/uploads/versions/diagram1---x----750-1087x---.jpg

OTHER CONCEPTS FOR DATA MANAGEMENT

Goal for today

- Data Warehouse
- Data Lake
- Data Mesh
- Data Fabric

-> What is the main difference between those and when to apply which

Data Lake vs Data Warehouse

	Data Warehouse	Data Lake
Type	processed	raw
Purpose	ready for analysis	open
Users	business users	data scientists
Accessibility	costly	easy

Data Lake vs Data Warehouse

GWDG

DATA WAREHOUSE	vs.	DATA LAKE
structured, processed	DATA	structured / semi-structured / unstructured, raw
schema-on-write	PROCESSING	schema-on-read
expensive for large data volumes	STORAGE	designed for low-cost storage
less agile, fixed configuration	AGILITY	highly agile, configure and reconfigure as needed
mature	SECURITY	maturing
business professionals	USERS	data scientists et. al.

https://www.kdnuggets.com/2015/09/data-lake-vs-data-warehouse-key-differences.html

Data Lake vs Data Warehouse

Schema-on-write

- Know your schema before building the data base
- Fast results
- Not agile
- Structured
- SQL

Schema-on-read

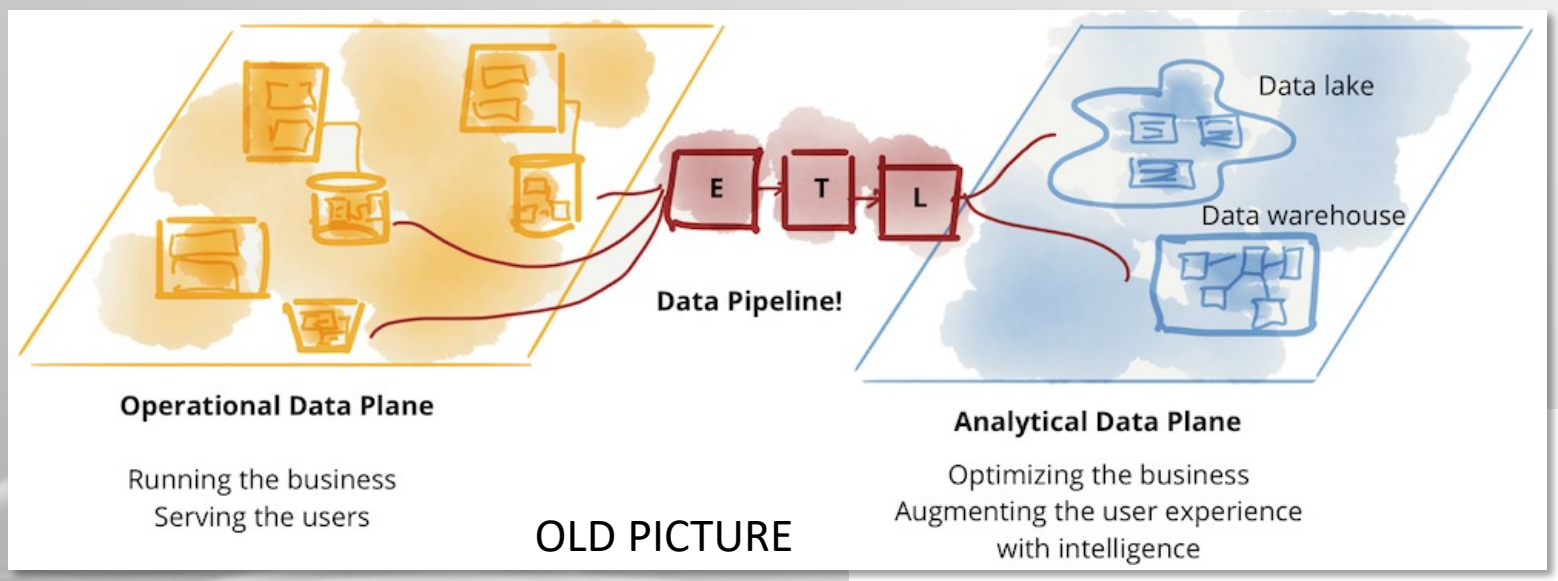
- Store raw data
- Slow results
- Very agile
- Unstructured
- NoSQL

Data Mesh

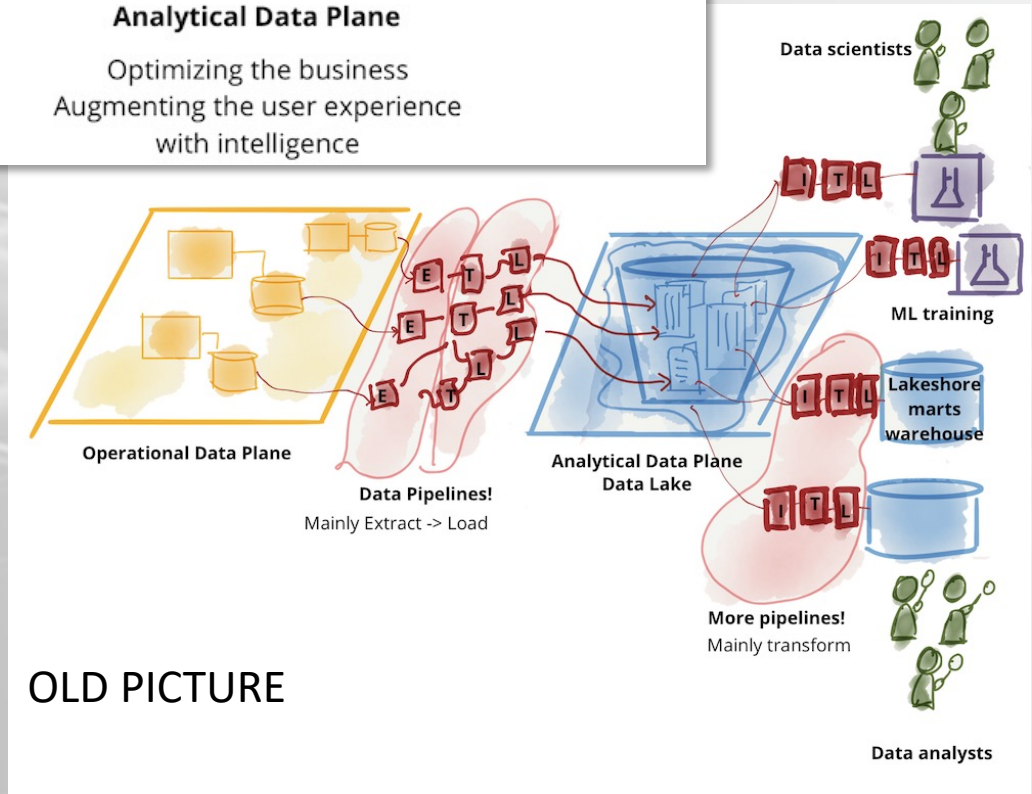
Zhamak Dehghani , founder of the idea

1) domain-oriented decentralized data ownership and architecture
   - derive domain specific policies and guidelines
2) data as a product
   - data producer are owner
3) self-serve data infrastructure as a platform
4) federated computational governance

Data Mesh by Dehghani

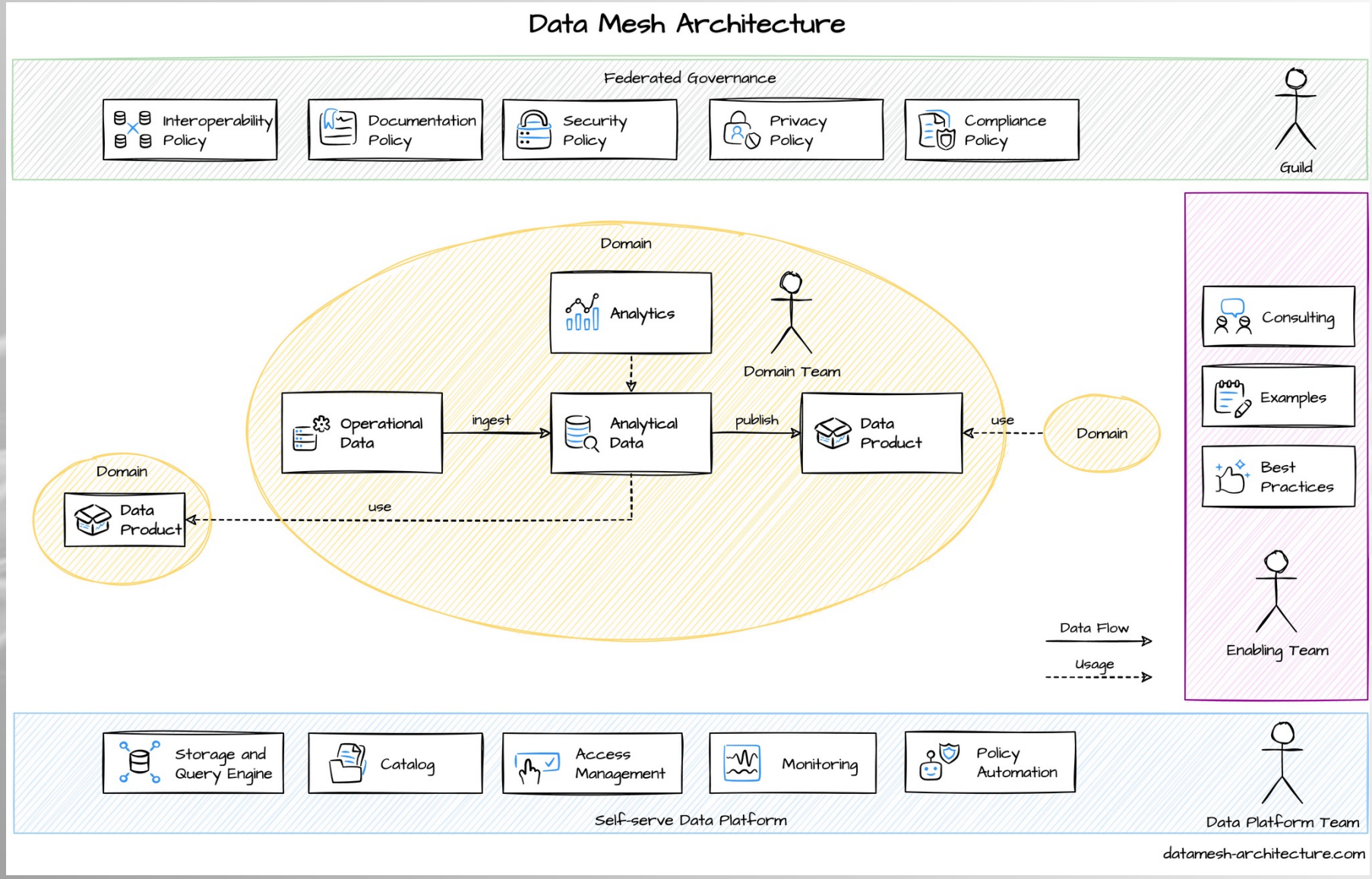


OLD PICTURE



OLD PICTURE

Data Mesh by Dehghani



Data Mesh vs. Data Lake

Data Mesh

- architecture for data governance
- data lakes can be used

Data Lake

- cheap storage architecture

Data Fabric

- an architectural concept
- focuses on automating data integration, data engineering and governance in a data value chain
- is based on the concept of "active metadata”
- uses knowledge graphs, semantics and AI/ML technology to identify patterns in different types of metadata
- use these insights to automate and orchestrate the data value chain
- complements a data mesh and is not mutually exclusive
- improves the data mesh by creating data products faster, enforcing global governance and making it easier to control the combination of multiple data products.

Data Fabric vs. Data Mesh

- Data Mesh
  - distributes responsibility among teams
  - promotes decentralised data management
- Data Fabric
  - provides a uniform view of the data
  - seamless integration across different systems

Cloud Solutions

GWDG

Data Warehouse

- Amazon Redshift
- Google BigQuery

Data Lakes

- Amazon S3
- Azure Blob Storage

ETL Tools

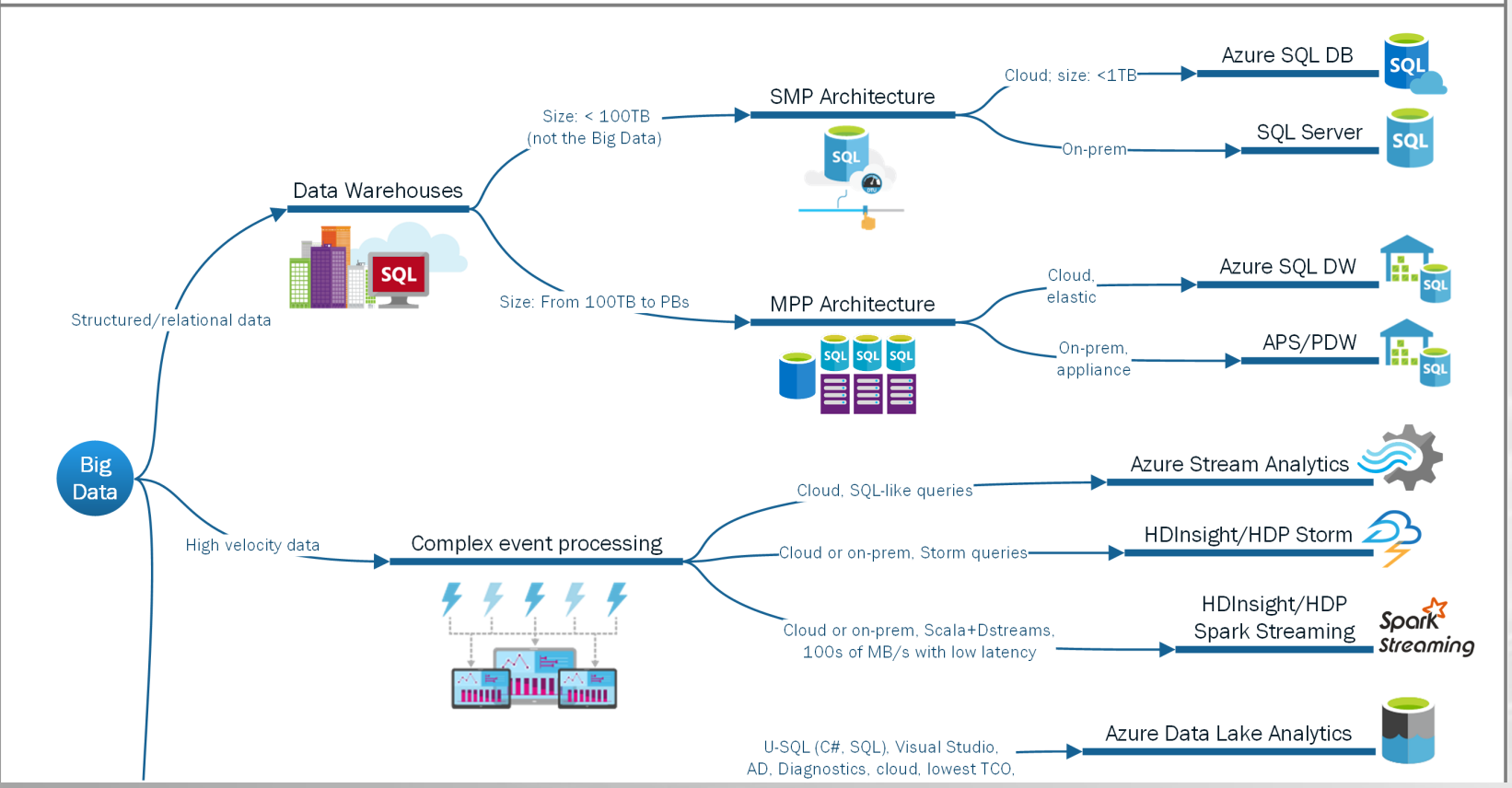
- Stitch
- Blendo
- Fivetran
- AWS Glue

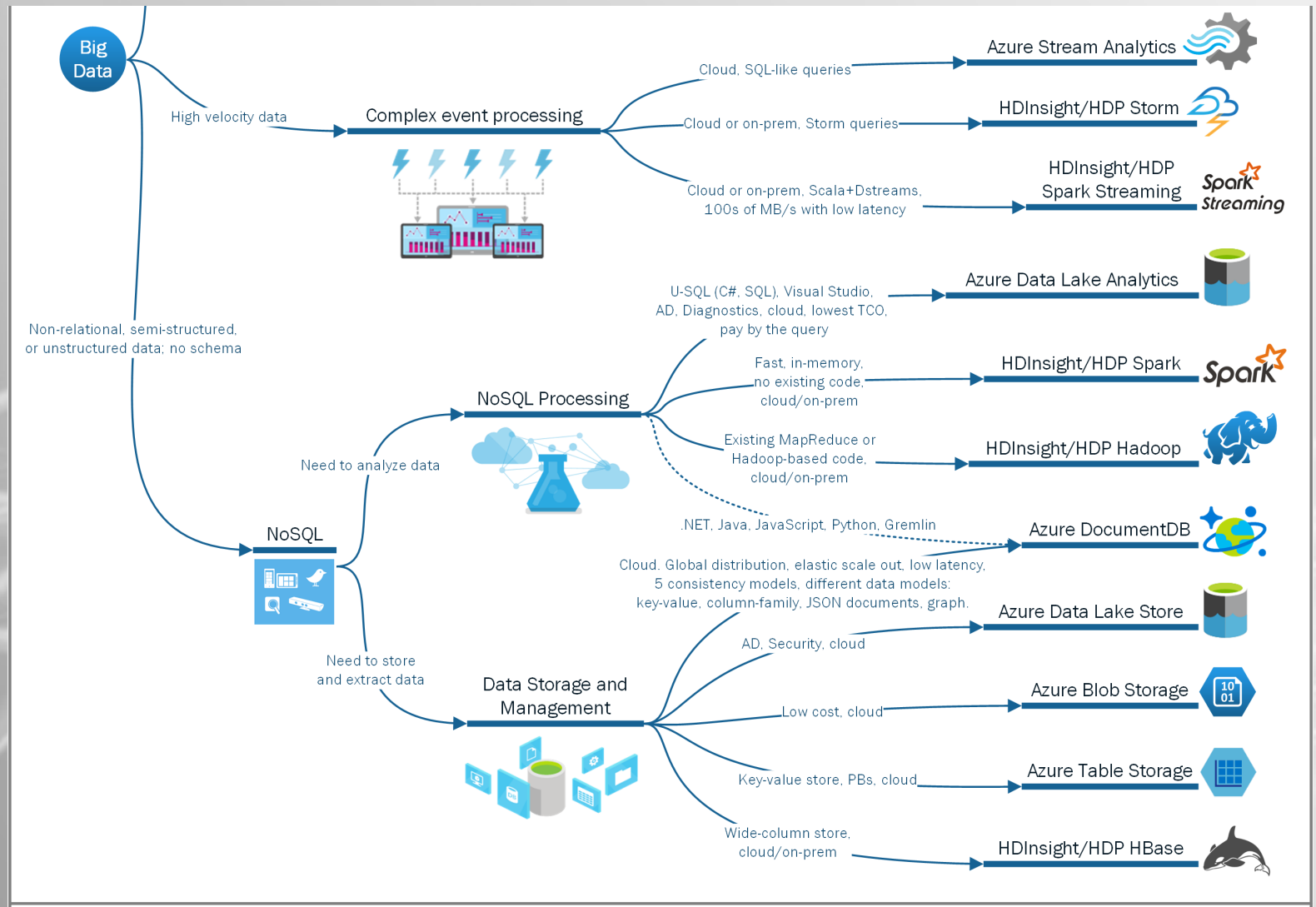
Analytics

- Tableau
- Qlik Sense
- Looker

Big Data Solutions Decision Tree

Version 2.01. May 30, 2016. Author: Ivan Kosyakov, Ph.D., Technical Architect in Microsoft Technology Center New York





Download Big Data Solutions Decision Tree at http://biz-excellence.com/2016/08/30/big-data-dt

2

METADATA

Metadata

- is a must
- connects data with data or services
- necessary for control and efficiency when processing data
- required for discovery and access

Problem

- meta data is distributed in different places
- no standard format is used

Meta Data Types

- Ownership and Golden Sources metadata
- Collaborative metadata
- Data quality metadata
- Business architecture metadata
- Technology metadata
- Semantics metadata
- BI and analytics metadata

- not all are always given
- use central meta data system

Meta Data

Meta Data should answer (partly):

- Who created it
- What is it
- When was it created
- How was it generated
- Where was it created
- How may it be used
- Licence

We need to put data into context!

Meta Data

Problem

- ”publication year” or ”pub year” or “Year”
- Missing standards
- Example standard: Dublin Core
  - <xs:element name="title" substitutionGroup="any"/>
  - <xs:element name="creator" substitutionGroup="any"/>
  - <xs:element name="subject" substitutionGroup="any"/>
  - <xs:element name="description" substitutionGroup="any"/>
  - <xs:element name="publisher" substitutionGroup="any"/>
  - <xs:element name="contributor" substitutionGroup="any"/>
  - <xs:element name="date" substitutionGroup="any"/>
  - <xs:element name="type" substitutionGroup="any"/>
  - <xs:element name="format" substitutionGroup="any"/>
  - <xs:element name="identifier" substitutionGroup="any"/>
  - <xs:element name="source" substitutionGroup="any"/>
  - <xs:element name="language" substitutionGroup="any"/>
  - <xs:element name="relation" substitutionGroup="any"/>
  - <xs:element name="coverage" substitutionGroup="any"/>
  - <xs:element name="rights" substitutionGroup="any"/>

https://www.dublincore.org/specifications/dublin-core/dcmi-terms/

Meta Data

- Example Publication as BibTex
  - author = {{Peter, H.} and {Bingert, S.} and {Klimchuk, J. A.} and {de Forest, C.} and {Cirtain, J. W.} and {Golub, L.} and {Winebarger, A. R.} and {Kobayashi, K.} and {Korreck, K. E.}},
  - title = {Structure of solar coronal loops: from miniature to large-scale },
  - DOI= "10.1051/0004-6361/201321826",
  - url= "https://doi.org/10.1051/0004-6361/201321826",
  - journal = {A\&A},
  - year = 2013,
  - volume = 556,
  - pages = "A104",
  - month = "",

Meta Data

- CERIF - Common European Research Information FormatCERIF is the standard that the EU recommends to its member states for recording information about research activity.
- Data PackageThe Data Package specification is a generic wrapper format for exchanging data, consisting of a folder containing data files and a descriptor file.
- DataCite Metadata SchemaA domain-agnostic list of core metadata properties chosen for the accurate and consistent identification of data for citation and retrieval purposes.
- DCAT - Data Catalog VocabularyDCAT is an RDF vocabulary designed to facilitate interoperability between data catalogs published on the Web.
- Dublin CoreA basic, domain-agnostic standard which can be easily understood and implemented, and as such is one of the best known and most widely used metadata standards.
- OAI-ORE - Open Archives Initiative Object Reuse and ExchangeDefines standards for the description and exchange of aggregations of Web resources.
- Observations and MeasurementsThis standard specifies an XML implementation for the OGC and ISO Observations and Measurements (O&M) conceptual model, including a schema for Sampling Features.
- PREMISThe PREMIS (Preservation Metadata: Implementation Strategies) Data Dictionary defines a set of metadata that most repositories of digital objects would need to record and use in order to preserve those objects over the long term.
- PROVA specification that provides a vocabulary to interchange provenance information.
- RDF Data Cube VocabularyThe Data Cube vocabulary is a core foundation which supports extension vocabularies to enable publication of other aspects of statistical data flows or other multi-dimensional data sets.
- Repository-Developed Metadata SchemasSome repositories have decided that current standards do not fit their metadata needs, and so have created their own requirements.

https://www.dcc.ac.uk/resources/subject-areas/general-research-data

Meta Data

- Example Data Package (given as Json-Schema)

```
profile
name
id
title
    propertyOrder : 40
    title : "Title"
    description : "A human-readable title."
    type : "string"
    examples
description
homepage
created
contributors
keywords
image
licenses
resources
sources
```

Schema

- a set of detailed guidelines that describe explicit rules and relationships for each metadata element in a standard
- is much more technical in nature than standards and provide definitions and guidance focusing on semantics and syntax

https://specs.frictionlessdata.io/data-package/#language

Meta Data

- With given meta data the described data can be interoperable

- FAIR principles
  - Findable
  - Accessable
  - Interoperable
  - Reusable

- The principles emphasise machine-actionability … <> … because humans increasingly rely on computational support to deal with data as a result of the increase in volume, complexity, and creation speed of data.

https://www.go-fair.org/fair-principles/

Consequences – FAIR

Findable

F1. (meta)data are assigned a globally **unique and eternally persistent identifier**

F2. data are described with **rich metadata**

F3. (meta)data are registered or indexed in a **searchable resource**

F4. metadata **specify** the data identifier

https://www.force11.org/group/fairgroup/fairprinciples

Consequences – FAIR

Accessible

A1. (meta)data are **retrievable by their identifier** using a **standardized communications protocol**

A1.1. the **protocol** is open, free, and universally implementable

A1.2. the **protocol** allows for an authentication and authorization procedure, where necessary

A2. **metadata are accessible**, even when the data are no longer available

https://www.force11.org/group/fairgroup/fairprinciples

Consequences – FAIR

Interoperable

I1. (meta)data use a **formal**, accessible, shared, and **broadly applicable language** for knowledge representation

I2. (meta)data use **vocabularies** that follow FAIR principles

I3. (meta)data include **qualified references** to other (meta)data

https://www.force11.org/group/fairgroup/fairprinciples

Consequences – FAIR

Re-usable

R1. meta(data) have a plurality of **accurate and relevant attributes**

R1.1. (meta)data are released with a clear and accessible **data usage license**

R1.2. (meta)data are associated with their **provenance**

R1.3. (meta)data **meet** domain-relevant **community standards**

https://www.force11.org/group/fairgroup/fairprinciples